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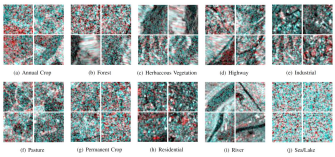
Quantum Neural Architecture Search for Satellite Image Classification

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94.0%	**31k**	**~800×**	**220**
EuroSAT accuracy of the best hybrid (mean of 3 seeds)	trainable parameters in that model	fewer parameters than ResNet-50 (98.6%, ImageNet-pretrained)	architectures evaluated out of 653,184

Motivation and Objectives

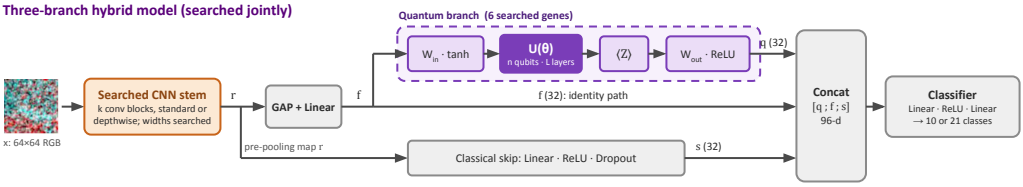
- Searching the influence of quantum machine learning on satellite data: EuroSAT (10 classes, 27,000 Sentinel-2 images, 64 px) and UC Merced (21 classes, 2,100 aerial images) serve as testbeds.
- A qubit is naturally a linear combination of basis states, so a parametrized quantum circuit is a natural drop-in layer for a neural network.
- With a fixed 9.5M-parameter CNN, the circuit was <0.1% of the model and its effect could not be measured, so we now search the CNN too, under a 50k-parameter cap.
- Goal: let NSGA-II design the CNN stem *and* the quantum circuit together, optimizing three objectives at once:
- **minimize total parameters · minimize transpiled 2-qubit gates · maximize accuracy**
- The outcome is a Pareto front of trade-offs between size, hardware cost, and accuracy, rather than a single best circuit.



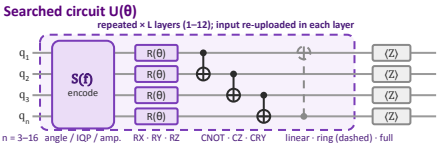
Not "what is our best accuracy?" but: does the quantum circuit beat the same model with the circuit removed?

Methodology

Three-branch hybrid model (searched jointly)



Searched circuit U(θ)



Genome: 9 searched genes

qubits	layers	embedding	rotation	entangler	topology
3 – 16	1 – 12	angle · IQP · amp.	RX · RY · RZ	CNOT · CZ · CRY	lin. · ring · full

14 × 12 × 3 × 3 × 3 × 3 = 13,608 quantum configurations

channels	hidden width	block type
6 width schedules	32 · 64 · 96 · 128	standard · depthwise

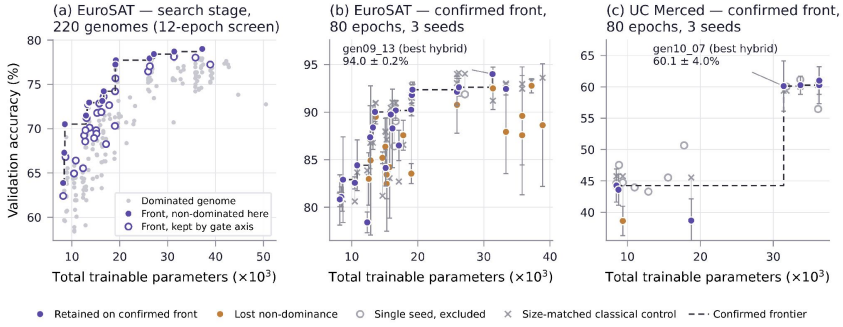
6 × 4 × 2 = 48 classical configurations

13,608 × 48 = 653,184 genomes; NSGA-II evaluates 220 of them (0.034%)

Search. NSGA-II, population 20 × 10 generations = 220 evaluations. Infeasible genomes are rejected before training (50k classical-parameter cap, qubit cap, gradient check). A 12-epoch screen drives the search; the front is retrained for 80 epochs × 3 seeds.

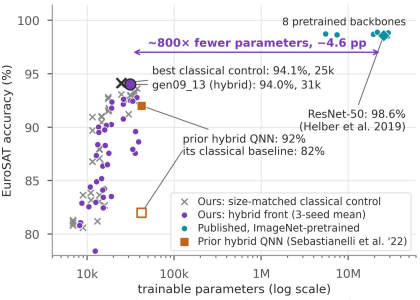
Size-matched control. Every front architecture is retrained with its quantum branch removed (same stem, data and budget). Δ = hybrid − control isolates what the circuit adds.

Results



(a) all 220 screened genomes (grey) and the search front. (b, c) front architectures retrained for 80 epochs × 3 seeds, mean ± SD: purple = still non-dominated, orange = lost non-dominance after retraining, grey × = size-matched classical control. UC Merced: chance level 4.8% (21 classes), 80 training images per class at 64×64.

EuroSAT: our front vs published models



Published: Helber et al. 2019 (ResNet-50); 8 backbones from a 2024 benchmark (inputs resized to 256×256); Sebastianelli et al. 2022. Ours: trained from scratch at 64×64, validation accuracy.

Main findings

- **Same accuracy class, far smaller.** gen09_13 reaches 94.0% ± 0.2% (3 seeds) with 31k parameters, ~800× fewer than ResNet-50 (98.6%, ImageNet-pretrained), trained from scratch on 64×64 images. Its size-matched classical control is just as small: the saving comes from the searched stem.
- **The quantum branch does not pay on average.** It beats its own size-matched control on 12 of 41 fully-seeded EuroSAT architectures (mean Δ = −1.0 pp). gen09_13 (+1.4 pp) is the exception and still needs replication.
- **Strong baselines matter.** A prior hybrid reported 92% against an 82% classical baseline at 42k parameters; here a 25k-parameter classical control alone reaches 94.1%.
- **Circuit cost buys nothing.** Accuracy follows parameters (ρ = 0.77), not transpiled 2-qubit gates (ρ = 0.24); the best circuit uses 7 qubits and 1 layer. The 12-epoch screen ranks well (ρ = 0.81), but only 54% of the front stays non-dominated after retraining.
- **UC Merced, a data-scarcity stress test.** 60.1% ± 4.0% (12.6× chance) with 80 training images per class; Δ = +0.7 pp. With 16× less data the whole front drops ~30 pp but keeps its shape.

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